

Fourier series

GNU Radio activity 3

Build square and triangle wave approximations by combining six sine waves. Then explore how their amplitudes and frequencies affect the result.

This activity prepares you to use the fast Fourier transform in Activity 4.

Square wave

1. Open your six-source signal generator from Activity 2.
2. Save it with a new filename.
3. Run the program. Enter the frequencies and amplitudes below.

Signal source	Frequency (Hz)	Amplitude
1	100	1.000
2	300	0.333
3	500	0.200
4	700	0.143
5	900	0.111
6	1100	0.091

Observe the Time Domain Display. The combined signal should resemble a square wave.

Triangle wave

4. In each amplitude QT GUI Range block, change the Start value from 0 to -1.
5. Enter the frequencies and amplitudes below.

Signal source	Frequency (Hz)	Amplitude
1	100	-1.000
2	300	0.111
3	500	-0.040
4	700	0.020
5	900	-0.012
6	1100	0.008

Run the program and observe the Time Domain Display. The combined signal should resemble a triangle wave.

Discussion

A Fourier series represents a periodic function as a sum of sine and cosine terms. These activities use six terms to approximate familiar waveforms.

The square wave approximation is not an exact square wave. Adding more terms improves the approximation, although overshoot remains near a discontinuity.

Signal processing often reverses this process. We start with measured samples and determine their frequency content using a discrete Fourier transform.

The fast Fourier transform, or FFT, computes that transform efficiently. Activity 4 uses an FFT block with your six-source generator.

Explore

Adjust the amplitudes and frequencies to create different wave patterns. Compare the resulting time and frequency displays.